IDR305: Introduction to Economics for Sport Management

Final Exam

Instructions (read carefully)

3 hours and 30 minutes, including all practical and technical actions needed to upload your answers.

All supporting materials permitted. It is not allowed to cooperate with or receive help from others.

Justify all your answers with economic concepts, logical reasoning, and love for sport.

Good Luck!

Question 1 (15 points)

Suppose there is a country that can produce two goods: Basketballs (X) and Footballs (Y). The country has 200 machines and 400 workers. Each machine can produce 5 units of X or 2 units of Y. Each worker can produce 2 units of X or 4 units of Y.

a. Draw the production possibilities frontier (PPF) of this country. Mark the point(s) where the curve changes slope, as well as the maximal production of X and of Y. (5 points)

b. Suppose the country needs to consume 1200 units of Y. How many units of X will it consume? (3 points)

c. Suppose there is now world trade, such that 1 unit of X is worth 1.5 units of Y. How many units of X and Y should the country produce, and why? (4 points)

d. Suppose that, during world trade, the country still needs to consume the number of units of X you found in (b). How many units of Y will it consume? (3 points)

Question 2 (15 points)

You are given the following functions:

1) Demand function: QD = 20 − 2p

2) Supply function: QS = 3p

where p is the price of a good in US dollars ($), and QD and QS are the quantity demanded and supplied, respectively.

a. What are the price and quantity in equilibrium? (4 points)

b. Show on a graph the cases where p=0, Q=0, and the price and quantity in equilibrium. (6 points)

c. What is the producer surplus? (2 points)

d. What is the consumer surplus? (2 points)

e. What is the total social welfare? (1 point)

Question 3 (15 points)

You are given the following firm's production function (one unit of labor (L) costs 10$):

TP (Q)	L	MP	FC	VC	TC	MC	ATC	AVC
0	0		40					
1	4							
2	7							
3	11							
4	16							
5	23							

a. Complete the table. (7 points)

b. What is the minimal price at which the firm will produce in the short term? (2 points)

c. What is the minimal price at which the firm will produce in the long term? (2 points)

d. It is known that each unit of production can be sold in the market for $70 (p=70). What would be the production of the firm? (2 points)

e. What would be its profit in the short term? (1 point)

f. What would be its profit in the long term? (1 point)

Question 4 (10 points)

a. Show on a graph the effects of a price floor. (5 points)

b. Provide a sport-related policy that embodies the concept of a price floor. (3 points)

c. How is this sport-related policy similar to, and different from, an ordinary price floor (e.g. minimum wage)? (2 points)

Question 5 (10 points)

There is a market with 9 workers who should be allocated between two fields, given the following production functions:

Field A

L	**TP**
0	0
1	6
2	11
3	15
4	18
5	20
6	21

Field B

L	**TP**
0	0
1	5
2	9
3	12
4	14
5	15
6	15

a. How many workers would work in each field if the price of one produced unit in Field A is $3 and the price of one produced unit in Field B is $2? (6 points)

b. What would be the wage in this market? (4 points)

Question 6 (10 points)

You have 1000 NOK that you can invest now under one of three saving options, to be withdrawn at the end of year 4. The annual interest rates are: 6% in year 1, 6% in year 2, 8% in year 3, and 9% in year 4.

Option 1: Invest the full 1000 NOK now.

Option 2: Invest 500 NOK now and 500 NOK at the end of year 2.

Option 3: Invest the full 1000 NOK at the end of year 2.

Which option yields the highest Future Value at the end of year 4? Show your calculations for all three options.

Question 7 (10 points)

a. In the NBA draft lottery, 4 balls are drawn (without replacement, order does not matter) from 14 numbered balls. Show the calculation for how many combinations exist, and explain why only 1,000 of them are actually assigned to teams. (4 points)

b. A team is assigned 70 of those 1,000 combinations. What is its probability of winning the first pick? (1 point)

c. Explain the difference between the NBA's Salary Cap and its Tax Level. What happens to a team in each of the three resulting zones? (5 points)

Question 8 (8 points)

a. What is UEFA's Squad Cost Ratio (SCR), and what spending does it cap? (3 points)

b. A football club has revenue of €300 million and spends €240 million combined on wages, transfer amortization, and agent fees. Is this club compliant with the permanent (2025/26) SCR threshold of 70%? Show your work. (3 points)

c. Briefly explain one criticism of FFP's break-even rule related to competitive balance. (2 points)

Question 9 (7 points)

a. What is monopsony power, and how did it apply to football clubs before the 1995 Bosman ruling? (4 points)

b. After the Bosman ruling removed this monopsony power, would you expect player wages to rise, fall, or stay the same? Briefly explain why. (3 points)